Supplementary Material for “Coverage-Qualified Evaluation-Contract Auditing for Aerial Vehicle Detection”

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TABLE S1
REGISTERED EVALUATION CONTRACTS. G IS ALL VALID GT, G_t THE SUPPORT-RETAINED TARGET, $D_t = G \setminus G_t$, AND H SOURCE IGNORE.

ID	Scored valid GT	Neutral regions	Step meaning
A	G	none	all-valid reference
F	G_t	none	removed objects are background
R	G_t	D_t	neutralize removed objects
S	G_t	$D_t \cup H$	add source ignores

TABLE S2
EXACT TP/FP/FN TRANSITION LEDGER AT CONFIDENCE AND IOU .25. A = ABSOLUTE 24 PX; N = NORMALIZED .015. AI-TOD IS CONDITIONED ON ITS 1,869 COVERED IMAGES.

Source	Rule	Step	ΔTP	ΔFP	ΔFN	Δneutral	ΔF_1
VisDrone	A	$A \to F$	-2256	$+2256$	-2288	0	$-.0360$
		$F \to R$	0	-2320	0	$+2320$	$+.0667$
		$R \to S$	0	-293	0	$+293$	$+.0093$
VisDrone	N	$A \to F$	-399	$+399$	-1344	0	$+.0178$
		$F \to R$	0	-356	0	$+356$	$+.0093$
		$R \to S$	0	-294	0	$+294$	$+.0078$
UAVDT	A	$A \to F$	-301	$+301$	-237	0	$-.0085$
		$F \to R$	0	-547	0	$+547$	$+.0038$
		$R \to S$	0	0	0	0	$.0000$
UAVDT	N	$A \to F$	-105	$+105$	-182	0	$-.0023$
		$F \to R$	0	-193	0	$+193$	$+.0014$
		$R \to S$	0	0	0	0	$.0000$
AI-TOD	A	$A \to F$	-11977	$+11977$	-2890	0	$-.3303$
		$F \to R$	0	-30232	0	$+30232$	$+.0800$
		$R \to S$	0	0	0	0	$.0000$
AI-TOD	N	$A \to F$	-5707	$+5707$	-1840	0	$-.1359$
		$F \to R$	0	-12587	0	$+12587$	$+.0660$
		$R \to S$	0	0	0	0	$.0000$

TABLE S3
AI-TOD COVERED–NONCOVERED IMAGE QUALIFICATION. BRACKETS ARE MEDIAN [Q1,Q3]. SMD USES COVERED MINUS NONCOVERED MEANS; W_1 IS IN THE FEATURE'S NATIVE UNIT.

Feature	Covered	Noncovered	SMD	W_1
Boxes/image	0 [0,1]	21 [7,58.5]	$-.673$	36.951
Median side (px)[a]	17 [12,22]	16 [14,18]	$+.448$	5.597
Fraction < 24 px[a]	.895 [.600,1]	.961 [.857,1]	$-.640$	.159
Fraction $< .015$[a]	.018 [0,.410]	.009 [0,.111]	$+.610$	.148
Nearest distance[b]	.031 [.016,.065]	.027 [.018,.055]	$+.029$	.009

[a] Nonempty images. [b] Images with at least two vehicle boxes; centers normalized by image size.

TABLE S4
COMPLETE POOLED RETAINED VALID VEHICLE GT (%) BEFORE PREDICTION MATCHING.

Rule	Threshold	VisDrone	UAVDT	AI-TOD
Absolute	16 px	85.7	99.6	47.6
	20 px	79.5	98.8	20.4
	24 px	73.3	95.1	9.7
	28 px	67.5	77.5	5.8
	32 px	62.1	56.4	4.2
	40 px	52.3	34.3	2.5
	48 px	42.7	30.0	1.4
	64 px	27.7	17.7	0.4
Normalized	.0050	100.0	100.0	99.9
	.0075	98.8	100.0	99.6
	.0100	96.5	99.8	98.2
	.0150	89.8	97.4	81.0
	.0200	82.4	80.0	47.6
	.0300	67.6	46.4	9.7

S1. Registered Contracts and Count Ledger

The four contracts alter one evaluator factor at a time. All detections are score ordered; retained valid GT is matched before an otherwise unmatched detection can be neutralized. Removed and source-ignore regions use intersection over prediction area at .5. Table S1 distinguishes the scored target from neutral regions explicitly.

Table S2 is the support-conditioned count-transition ledger behind Fig. 2 of the main paper. Every row is an exact adjacent-contract difference for one fixed prediction artifact. Summing the three ΔF_1 values gives $F_{1,S} - F_{1,A}$ to machine precision. Because F_1 is nonlinear, the values are path-specific transitions rather than additive causal effects that are invariant to reordering.

S2. Coverage Qualification and Full Support Grid

Raw prediction names establish coverage before class filtering. VisDrone and UAVDT cover all declared images. AI-TOD covers 1,869/2,804 images and is therefore eligible for full-annotation support diagnosis, a covered-subset metric replay, but not full-benchmark metric or rank inference. Its 1,869 covered images contain 16,900 boxes; the 935 noncovered images contain 43,004.

The image-level record shows that incomplete coverage is not exchangeable with the full validation set. The covered subset has many empty vehicle images, yet among nonempty images its per-image support composition also differs. At the box level, covered and noncovered median max sides are 12 and 16 pixels; the normalized-support Wasserstein distance is .00370. No weighting or imputation is used to extrapolate the conditional metric endpoints.

S3. Paired Rank Records and Evaluator Controls

Table S5 gives the localization-by-metric control for the Y11m–ASD pair. Sequence clusters are the leading seven-digit VisDrone filename tokens (76 clusters). Every replicate includes all images in each sampled sequence, applies identical multiplicities to both configurations, and recomputes global

TABLE S5
Y11M–ASD SEQUENCE-PAIRED DIFFERENCES UNDER TERMINAL CONTRACT S (10,000 REPLICATES). A = 24 PX; N = .015. IOU IS THE VALID-GT MATCH THRESHOLD; NEUTRALIZATION REMAINS AT INTERSECTION-OVER-PREDICTION .5.

Rule	IoU/metric	Y11m	ASD	Diff.	Seq. 95%
A	.25/F_1 $(c = .25)$	.8927	.8697	$+.0229$	[.0185,.0275]
	.25/max-F_1	.8979	.8816	$+.0163$	[.0119,.0213]
	.25/AP	.9453	.9372	$+.0081$	[.0051,.0120]
	.50/F_1 $(c = .25)$	.8799	.8577	$+.0222$	[.0180,.0269]
	.50/max-F_1	.8886	.8745	$+.0141$	[.0099,.0191]
	.50/AP	.9231	.9217	$+.0013$	[$-.0033$,.0052]
N	.25/F_1 $(c = .25)$	.8831	.8673	$+.0158$	[.0118,.0201]
	.25/max-F_1	.8831	.8732	$+.0099$	[.0060,.0140]
	.25/AP	.9310	.9315	$-.0005$	[$-.0030$,.0047]
	.50/F_1 $(c = .25)$	.8664	.8524	$+.0141$	[.0104,.0181]
	.50/max-F_1	.8692	.8639	$+.0054$	[.0013,.0103]
	.50/AP	.8986	.9096	$-.0110$	[$-.0181$,$-.0057$]

counts or pooled-detection AP. AP is never averaged across images or sequences.

Fixed- and optimized-confidence F_1 favor Y11m at both localization thresholds. At IoU .25, AP favors Y11m for A and is unresolved for N; at IoU .50, AP is unresolved for A and favors ASD for N. A sequence-resolved ASD-favoring normalized conclusion therefore occurs only in the cell combining stricter localization with score integration. Image-paired controls preserve these endpoint directions and remain in the released records.

The AP cap is 2,000 detections per image. The observed maximum is 1,881, so no pool image is truncated; caps of 100 and 300 alter dense-image inputs. The independent AP implementation and COCOeval agree exactly for AP50 (maximum absolute difference $< 10^{-12}$). AP25 keeps the S-contract neutralization threshold fixed at .5, whereas COCOeval couples crowd matching to the evaluated IoU; COCOeval equivalence is therefore not asserted for that deliberately controlled cell. A threshold-first Hungarian replay selects the same top configuration as greedy matching in 24/24 settings; the maximum absolute F_1 difference is .00546.

The full 14-support-by-eight-confidence rank surface remains exploratory because its small-gap cells were identified after inspecting the prespecified grid. No pointwise interval from a selected boundary cell is used as confirmatory evidence. Machine-readable full surfaces, matching controls, maxDets diagnostics, and DOTA-v2 structural support distances remain in the fixed repository release.

S4. Ancillary Qualification Controls

Figure 1 separates two boundary checks that precede metric interpretation. The AI-TOD density gap is visible at the image level, while the box-level curves show that coverage selection also changes the support composition of the observed target. The cap panel uses mapped vehicle predictions before AP truncation: a low cap would modify the prediction set for every configuration by a different amount. The declared cap of 2,000 exceeds the observed per-image maximum of 1,881 and therefore preserves every mapped prediction.

Figure 2 provides the structural context omitted from the main figures. Normalization changes the coordinate system and greatly compresses the numerical scale, but it does not collapse the four empirical distributions. In particular, the largest normalized pairwise distance is .031 between VisDrone and AI-TOD, whereas VisDrone and DOTA-v2 differ by .006. These are distributional distances, not annotation-quality scores, and no detector result is inferred from them.

TABLE S6
MINIMAL REPLAY CHAIN IN THE RELEASED REPOSITORY.

Audit object	Released record or script
Coverage gate	`outputs/coverage_qualification/` and `evaluate_aitod_coverage_qualification.py`
Support denominator	`outputs/scale/box_scale_records.parquet` and raw-coverage manifests
Contract path	`outputs/contract_path/`, `contract_path.py`, and `evaluate_contract_path_headlines.py`
Metric-qualified rank	`outputs/metric_qualified_rank/` and `evaluate_metric_qualified_visdrone_rank.py`
Metric-factorial control	`outputs/metric_factorial_control/` and `evaluate_metric_factorial_control.py`
Figures and tests	Four `build_*.py` scripts; contract, support, matching, and AP regression tests
Repository	https://github.com/Marchematics/aerial-evaluation-audit

S5. Reproducibility Record

The data-free release provides configurations, derived records, scripts, and SHA-256 checksums. Licensed imagery, trained weights, and raw prediction caches remain external and are referenced by manifest path and hash. Reproduction order is coverage $\rightarrow$ support $\rightarrow$ contract endpoints/ledger $\rightarrow$ paired metric qualification $\rightarrow$ figures.

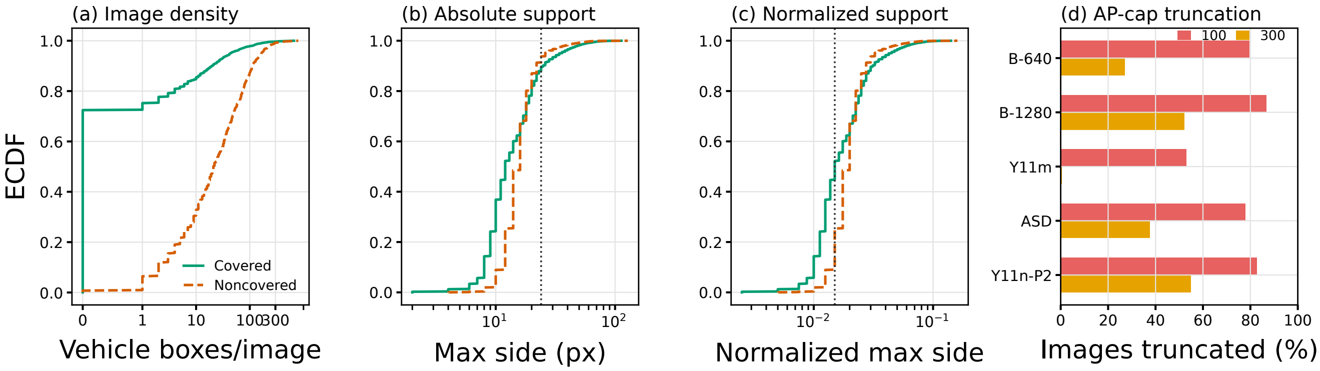

Fig. 1. Ancillary qualification records. (a)–(c) AI-TOD covered and noncovered image groups differ in density and support; vertical lines mark 24 px and .015. ECDFs in (b) and (c) use individual vehicle boxes. (d) Fractions of the 548 common VisDrone images whose mapped prediction counts exceed maxDets 100 or 300. At maxDets 2000, all five fractions are zero.

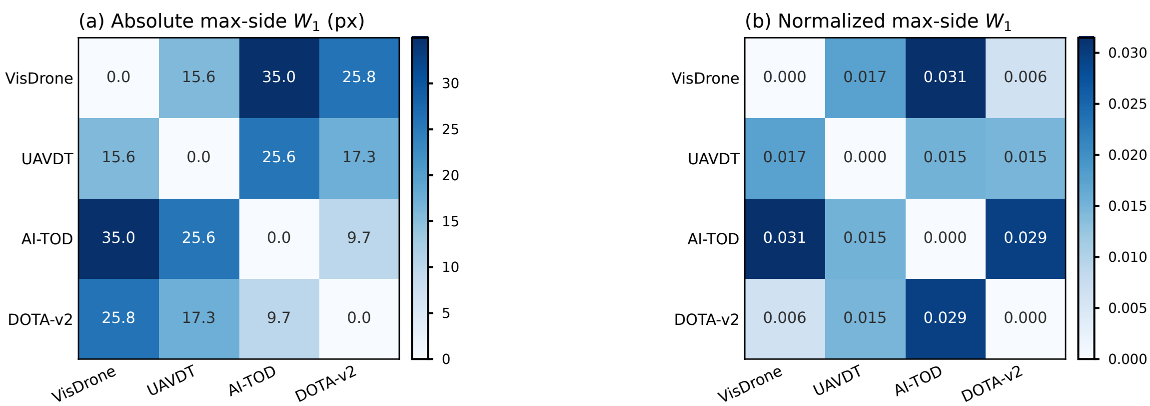

Fig. 2. Pairwise empirical Wasserstein distances for full-annotation vehicle support. (a) Absolute max side is scale covariant. (b) Normalized max side is resize invariant but retains nonzero source distances. DOTA-v2 is structural-only evidence and does not enter prediction-conditioned metric or rank claims.

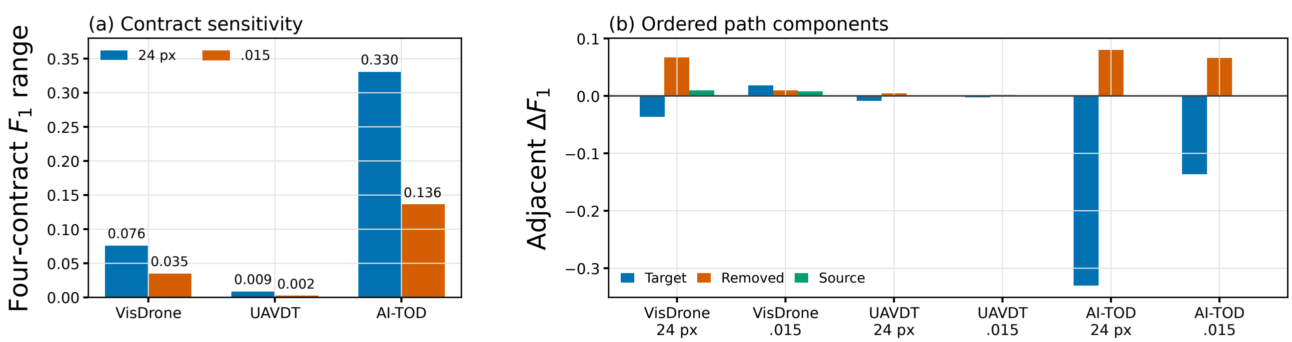

Fig. 3. Cross-source summary of the declared headline path. (a) Four-contract micro-F_1 ranges. (b) Adjacent path changes, preserving the declared order rather than treating them as order-free causal components. Normalized .015 reduces the range in all three sources but leaves source-dependent residual sensitivity.